

ENLANG FOR DEVELOPERS

The Definitive Guide to Building Full-Stack Software with Natural English

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"Programming should be as clear, expressive, and frictionless as human thought. EnLang makes natural English a first-class, production-grade programming language across the entire software stack."

— Spandan Prayas Patra

TABLE OF CONTENTS

- Chapter 1: Introduction & Philosophical Architecture
- Chapter 2: Variables, Optional Types, Output & Expression Syntax
- Chapter 3: The 3-Level Native Interactivity Architecture
- Chapter 4: Comprehensive Loop Taxonomy (For, While, Repeat, Recursion)
- Chapter 5: Functions, Natural Declarations, Interfaces & Async / Await
- Chapter 6: Pattern Matching & Decision Engine
- Chapter 7: Exception Handling & Error Control
- Chapter 8: Built-in Collections — Lists, Arrays, Dictionaries, Sets
- Chapter 9: String Manipulations, Math & DateTime Primitives
- Chapter 10: File I/O, System Operations & Security
- Chapter 11: Frontend Structural Markup (`.enlgf` → HTML5)
- Chapter 12: Styling & Design Systems (`.enlgd` → CSS3)
- Chapter 13: Client-Side Scripting (`.enlgs` → JavaScript ES6+)
- Chapter 14: Database Schemas & Queries (`.enlgdb` → SQL)
- Chapter 15: Native NLP & AI Primitives
- Chapter 16: EnLang Web Server & Multi-File Architecture
- Chapter 17: Developer Tooling — Syntax Checker, Linter & Interactive Debugger
- Chapter 18: Canonical Grammar Rules, Order & Syntax Layout Specification
- Chapter 19: Complete Production Case Study — Lumina Workspace
- Appendix A: Universal Natural Syntax Cheatsheet
- Appendix B: CLI & EnLang Package Manager (EPM) Manual

CHAPTER 17: DEVELOPER TOOLING — SYNTAX CHECKER, LINTER & INTERACTIVE DEBUGGER

EnLang includes enterprise-grade developer tooling for static code analysis, linting, and step-by-step interactive debugging.

17.1 EnLang Static Syntax Checker & Linter (`enlang check`)

The `enlang check` command performs static analysis on EnLang source files without executing them.

```
enlang check main.enlg
```

Automated Linter Checks:

- **4-Space Indentation Rule:** Verifies block indentation alignment.
- **Block Header Colons:** Ensures block statements (`if`, `for`, `while`, `function`) terminate with `:`.
- **Block Closure Verification:** Detects unclosed `match` or `interface` blocks.
- **Unclosed Strings:** Identifies unescaped or missing quote pairs.
- **Ambiguous Phrase Warnings:** Warns about non-canonical phrases (e.g., `is bigger than`) and suggests canonical replacements.

Example Linter Diagnostic Output:

```
=====
EnLang Syntax Checker & Linter – bad_syntax_test.enlg
=====
[ERROR] Line 3: Unclosed string literal detected. (Suggestion: Ensure string quotes are closed properly)
[ERROR] Line 4: Block header missing trailing colon ':'. (Suggestion: Add a colon ':' at the end of the line)
[ERROR] Line 4: Unsupported natural phrase detected in statement. (Suggestion: Use 'is greater than' instead of 'is bigger than')
=====
Result: 3 Error(s), 0 Warning(s)
=====
```

17.2 EnLang Interactive CLI Debugger (`enlang debug`)

The `enlang debug` command launches an interactive step-by-step execution debugger with live variable inspection.

```
enlang debug app.enlg
```

Interactive Debugger Commands:

- **s** / **step**: Step to next line of EnLang code.
- **c** / **continue**: Resume execution until next breakpoint or termination.
- **v** / **vars**: Display live variable table and values.
- **b** **<N>** / **break** **<N>**: Set breakpoint at line number **N**.
- **e** **<expr>** / **eval** **<expr>**: Evaluate expression in live execution frame.
- **q** / **quit**: Exit debugger session.

APPENDIX B: CLI & PACKAGE MANAGER (EPM) MANUAL

```
enlang run app.enlg          # Compiles & executes backend logic
enlang check app.enlg        # Performs static analysis & linting
enlang debug app.enlg        # Launches interactive step-by-step debugger
enlang build index.enlgf     # Transpiles source into native target file
enlang server --port 8000    # Launches zero-config HTTP host server
enlang add py:requests       # Installs Python PyPI package
enlang add web:chart.js      # Installs Web NPM package
```

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